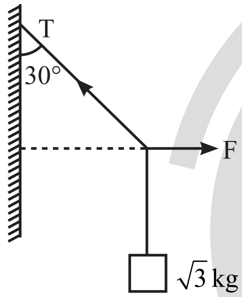
Newton's Laws of Motion

Single Correct Type Questions

1. A block of $\sqrt{3}$ kg is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30º with the wall. The tension T in the string is:
(Given $g = 10$ ms^{-2}) **[30 Jan, 2023 (Shift-II)]**



(*a*) 20 N (*b*) 25 N (*c*) 10 N (*d*) 15 N

2. At any instant the velocity of a particle of mass 500 g is $\left(2t\hat{i} + 3t^2\hat{j}\right)ms^{-1}$. If the force acting on the particle at $t = 1$s is $\left(\hat{i} + x\hat{j}\right)N$. Then the value of x will be: **[08 Apr, 2023 (Shift-1)]**

(*a*) 3 (*b*) 4 (*c*) 6 (*d*) 2

3. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): An electric fan continues to rotate for some time after the current is switched off.

Reason (R): Fan continuous to rotate due to inertia of motion.

In the light of above statements, choose the most appropriate answer from the options given below. **[10 Apr, 2023 (Shift-II)]**

(*a*) A is correct but R is not correct.

(*b*) Both A and R are correct and R is the correct explanation of A.

(*c*) A is not correct but R is correct.

(*d*) Both A and R are correct but R is not the correct explanation of A.

4. A particle of mass m is moving in the xy-plane such that its velocity at a point (x, y) is given as $\vec{v} = \alpha(y\hat{x} + 2x\hat{y})$, where α is a non-zero constant. What is the force $\vec{F}$ acting on the particle? **[JEE Adv, 2023]**

(*a*) $\vec{F} = 2m\alpha^2(x\hat{x} + y\hat{y})$

(*b*) $\vec{F} = m\alpha^2(y\hat{x} + 2x\hat{y})$

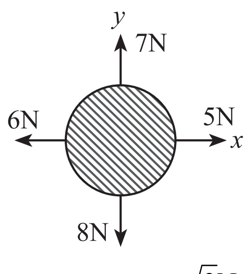
(*c*) $\vec{F} = 2m\alpha^2(y\hat{x} + x\hat{y})$

(*d*) $\vec{F} = m\alpha^2(x\hat{x} + 2y\hat{y})$

5. A mass of 10 kg is suspended by a rope of length 4m, from the ceiling. A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45º with the vertical. Then F equals
(Take $g = 10$ ms^{-2} and the rope to be massless) **[7 Jan, 2020 (Shift-II)]**

(*a*) 75 N (*b*) 90 N (*c*) 100 N (*d*) 70 N

6. For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero? **[25 July, 2022 (Shift-II)]**



(*a*) $\sqrt{2}$N, 45^o (*b*) $\sqrt{2}$N, 135^o

(*c*) $\frac{2}{\sqrt{3}}$N, 30^o (*d*) 2N, 45º

7. A monkey of mass 50 kg climbs on a rope which can withstand the tension (T) of 350 N. If monkey initially climbs down with an acceleration of 4 m/s^2 and then climbs up with an acceleration of 5 m/s^2. Choose the correct option
(Take $g = 10$ ms^{-2}) **[26 July, 2022 (Shift-I)]**

(*a*) $T = 700$ N while climbing upward

(*b*) $T = 350$ N while going downward

(*c*) Rope will break while climbing upward

(*d*) Rope will break while going downward

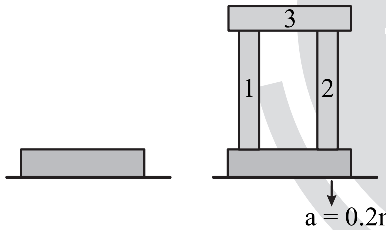
8. A force $\vec{F} = \left(40\hat{i} + 10\hat{j}\right)N$ acts on a body of mass 5 kg.

If the body starts from rest, its position vector $\vec{r}$ at time t =10s, will be: **[25 July, 2021 (Shift-II)]**

(*a*) $\left(100\hat{i} + 100\hat{j}\right)m$ (*b*) $\left(400\hat{i} + 100\hat{j}\right)m$

(*c*) $\left(400\hat{i} + 400\hat{j}\right)m$ (*d*) $\left(100\hat{i} + 400\hat{j}\right)m$

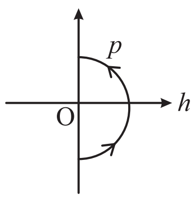
9. A steel block of 10 kg rests on a horizontal floor as shown. When three iron cylinders are placed on it as shown, the block and cylinders go down with an acceleration 0.2m/s^2. The normal reaction R' by the floor if mass of the iron cylinders are equal and of 20kg each, is ________N.
[Take g = 10m/s^2 and μ_s = 0.2] **[20 July, 2021 (Shift-I)]**

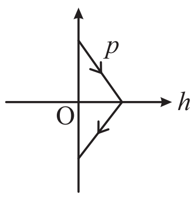


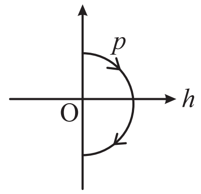
(*a*) 714 (*b*) 684 (*c*) 716 (*d*) 686

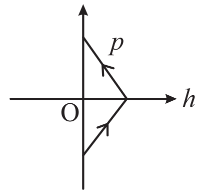
10. A ball is thrown vertically up (taken as +z-axis) from the ground. The correct momentum height (p-h) diagram is:

[9 April, 2019 (Shift-I)]

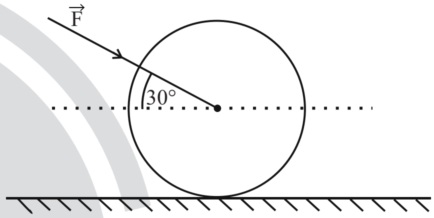
(*a*)


(*b*)


(*c*)


(*d*)


11. As shown in figure, a 70kg garden roller is pushed with a force of $\vec{F} = 200\,N$ at an angle of 30º with horizontal. The normal reaction on the roller is (Given $g = 10$ ms^{-2})

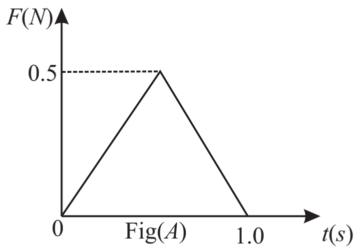


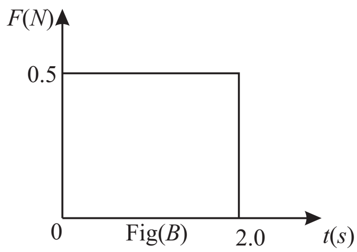
[31 Jan, 2023 (Shift-I)]

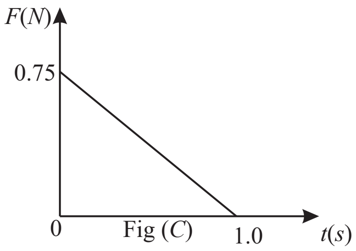
(*a*) $800\sqrt{2}\,N$ (*b*) $600\,N$

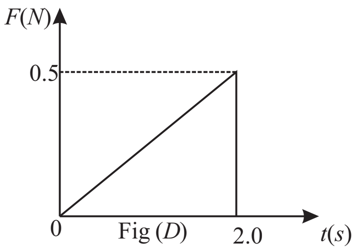
(*c*) $800\,N$ (*d*) $200\sqrt{3}\,N$

12. Figures (A), (B), (C) and (D) show variation of force with time.









The impulse is highest in figure.

[1 Feb, 2023 (Shift-II)]

(*a*) Fig (*C*)

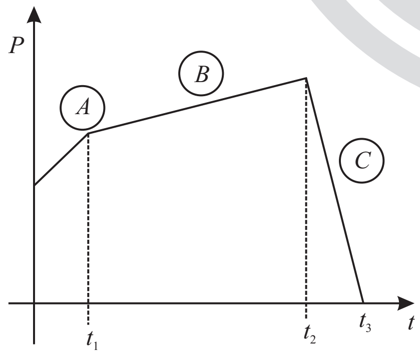
(*b*) Fig (*B*)

(*c*) Fig (*A*)

(*d*) Fig (*D*)

13. The figure represents the momentum time ($P - t$) curve for a particle moving along an axis under the influence of the force. Identify the regions on the graph where the magnitude of the force is maximum and minimum respectively?

If $(t_3 - t_2) < t_1$. **[30 Jan, 2023 (Shift-I)]**



(*a*) *C* and *A* (*b*) *B* and *C*

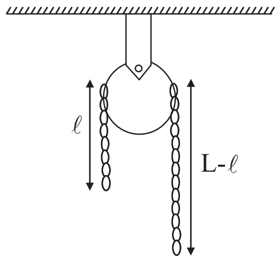
(*c*) *C* and *B* (*d*) *A* and *B*

14. 100 balls each of mass *m* moving with speed *v* simultaneously strike a wall normally and are reflected back with the same speed, in time *t s*. The total force exerted by the balls on the wall is **[31 Jan, 2023 (Shift-I)]**

(*a*) $\frac{100mv}{t}$ (*b*) $\frac{200mv}{t}$ (*c*) $200\ mvt$ (*d*) $\frac{mv}{100t}$

15. A uniform metal chain of mass *m* and length '*L*' passes over a massless and frictionless pulley. It is released from rest with a part of its length 'ℓ' is hanging on one side and rest of its length '$L - \ell$' is hanging on the other side of the pully. At a certain point of time, when $\ell = \frac{L}{x}$ the acceleration of the chain is $\frac{g}{2}$. The value of *x* is ________.

[28 July, 2022 (Shift-II)]



(*a*) 6

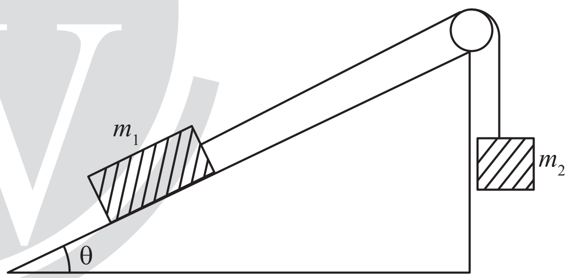
(*b*) 2

(*c*) 1.5

(*d*) 4

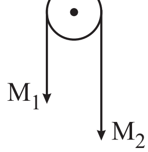
16. Two bodies of masses m_1 = 5kg and m_2 = 3kg are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be: [Take $g = 10\ ms^{-2}$]

[29 July, 2022 (Shift-II)]



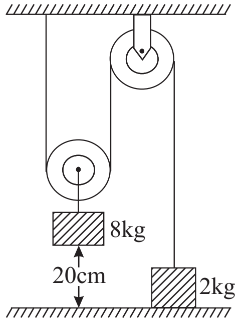
(*a*) 30 *N* (*b*) 40 *N* (*c*) 50 *N* (*d*) 60 *N*

17. Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1 the acceleration of the system is a_1. When the mass M_2 is thrice that of M_1. The acceleration of system is a_2. The ratio a_1/a_2 will be **[26 July, 2022 (Shift-II)]**



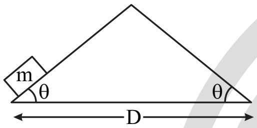
(*a*) $\frac{1}{3}$ (*b*) $\frac{2}{3}$ (*c*) $\frac{3}{2}$ (*d*) $\frac{1}{2}$

18. The boxes of masses 2 kg and 8 kg are connected by a massless string passing over smooth pulleys. Calculate the time taken by box of mass 8 kg to strike the ground starting from rest. (use $g = 10\ m/s^2$) **[27 Aug, 2021 (Shift-II)]**

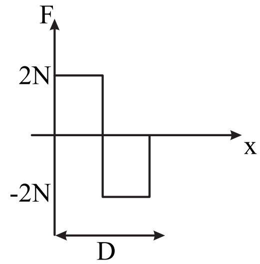


(*a*) 0.25 s (*b*) 0.4 s (*c*) 0.34 s (*d*) 0.2 s

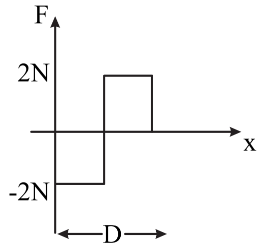
19. An object of mass 'm' is being moved with a constant velocity under the action of an applied force of $2N$ along a frictionless surface with following surface profile. The correct applied force vs distance graph will be: **[1 Sept, 2021 (Shift-II)]**



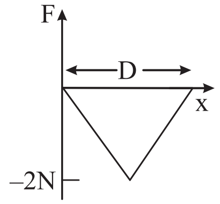
(*a*)



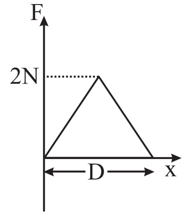
(*b*)



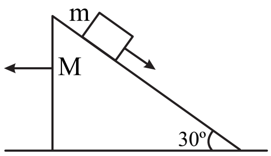
(*c*)



(*d*)



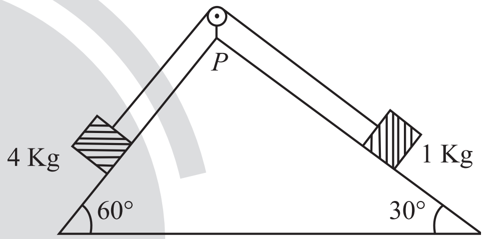
20. A block of mass m slides on the wooden wedge, which in turn slides backward on the horizontal surface. The acceleration of the block with respect to the wedge is: Given $m = 8$ kg, $M = 16$ kg Assume all the surfaces shown in the figure to be frictionless. **[1 Sept, 2021 (Shift-II)]**



(*a*) $\frac{6}{5}g$ (*b*) $\frac{3}{5}g$ (*c*) $\frac{2}{3}g$ (*d*) $\frac{4}{3}g$

21. As per given figure, a weightless pulley P is attached on a double inclined frictionless surface. The tension in the string (massless) will be (if $g = 10 m/s^2$)

[24 Jan, 2023 (Shift-I)]

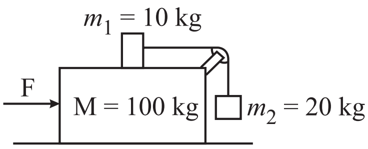


(*a*) $\left(4\sqrt{3}+1\right)N$ (*b*) $4\left(\sqrt{3}+1\right)N$

(*c*) $4\left(\sqrt{3}-1\right)N$ (*d*) $\left(4\sqrt{3}-1\right)N$

22. Three masses $M = 100$ kg, $m_1 = 10$ kg and $m_2 = 20$ kg are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of 2 ms^{-2}. The value of F is

(Take $g = 10\ ms^{-2}$) **[26 July, 2022 (Shift-I)]**



(*a*) 3360 N
(*b*) 3380 N
(*c*) 3120 N
(*d*) 3240 N

23. A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of 'a' will be: **[29 June, 2022 (Shift-II)]**

(a) $g/4$ (b) $g/2$ (c) $3g/4$ (d) g

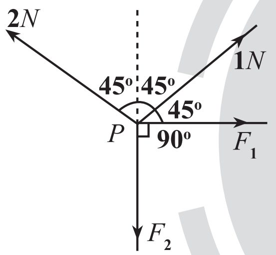
24. Force acts for 20 sec on a body of mass 20 kg, starting from rest, after which the force ceases and then the body describes 50 m in the next 10 s. The value of force will be: **[29 Jan, 2023 (Shift-II)]**

(a) 40 N (b) 5 N (c) 20 N (d) 10 N

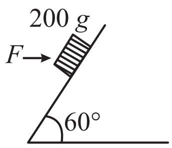
Integer Type Questions

25. A body of mass 2 kg moves under a force of $\left(2\hat{i}+3\hat{j}+5\hat{k}\right)N$. It starts from rest and was at the origin initially. After 4s, its new coordinates are (8, b, 20). The value of b is: (Round off to the Nearest Integer). **[16 March, 2021 (Shift-II)]**

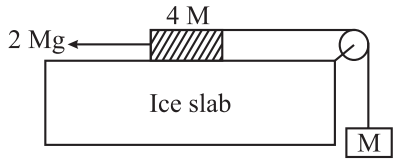
26. Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force F_1 to F_2 is 1: x where x = _________. **[25 July, 2022 (Shift-I)]**



27. A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force $F=\sqrt{x}N$ as shown in figure. The value of x = _______________. **[25 June, 2022 (Shift-II)]**



28. A hanging mass M is connected to a four times bigger mass by using a string pulley arrangement, as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by 2 Mg force. In this situation, tension in the string is x/5 Mg for x = _______. Neglect mass of the string and friction of the block (bigger mass) with ice slab. **[28 June, 2022 (Shift-I)]**

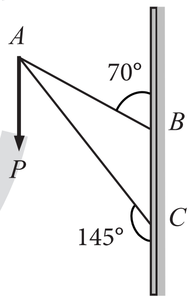


29. Consider a frame that is made up of two thin massless rods AB and AC as shown in the figure. A vertical force $\vec{P}$ of magnitude 100N is applied at point A of the frame. Suppose the force is $\vec{P}$ resolved parallel to the arms AB and AC of the frame. The magnitude of the resolved component along the arm AC is xN. **[16 March, 2021 (Shift-I)]**

The value of x, to the nearest integer, is ___________.

[Given : sin(35°) = 0.573, cos(35°) = 0.819

sin(110°) = 0.939, cos(110°) = –0.342]



30. A person standing on a balance inside a stationary lift measures 60 kg. The weight of that person if the lift descends with uniform downward acceleration of 1.8 m/s^2 will be ________ N. [g = 10 m/s^2]

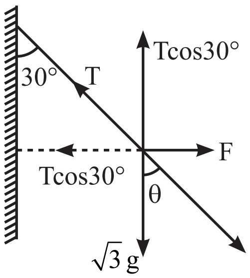
[26 Feb, 2021 (Shift-I)]

ANSWER KEY

1. (*a*)	**2.** (*a*)	**3.** (*b*)	**4.** (*a*)	**5.** (*c*)	**6.** (*a*)	**7.** (*c*)	**8.** (*b*)	**9.** (*d*)	**10.** (*c*)
11. (*c*)	**12.** (*b*)	**13.** (*c*)	**14.** (*b*)	**15.** (*d*)	**16.** (*b*)	**17.** (*b*)	**18.** (*b*)	**19.** (*b*)	**20.** (*c*)
21. (*b*)	**22.** (*a*)	**23.** (*c*)	**24.** (*b*)	**25.** [12]	**26.** [3]	**27.** [12]	**28.** [6]	**29.** [164]	**30.** [492]

EXPLANATIONS

1. (*a*)



Tcos30° = mg

$$\cos 30° = \frac{\sqrt{3}g}{T} \Rightarrow \frac{\sqrt{3}}{2} = \frac{\sqrt{3}g}{T}$$

$\Rightarrow T = 20\ N$

2. (*a*) $\vec{v} = 2t\hat{i} + 3t^2\hat{j}$

$$\vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i} + 6t\hat{j}$$

When $t = 1$ sec $\Rightarrow \vec{a} = 2\hat{i} + 6\hat{j}$

$$\therefore \vec{F} = m\vec{a} = 0.5\left(2\hat{i} + 6\hat{j}\right) = \hat{i} + 3\hat{j}$$

$$\hat{i} + x\hat{j} = i + 3\hat{j} \Rightarrow x = 3$$

3. (*b*) Fact based.

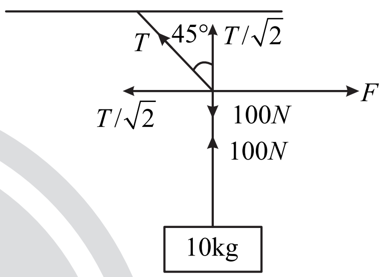
4. (*a*) Given, $\vec{a} = \frac{dv_x}{dt}\hat{x} + \frac{dv_y}{dt}\hat{y}$

$$\because \frac{dv_x}{dt} = \alpha\frac{dy}{dt} = 2\alpha^2 x$$

$$\frac{dv_y}{dt} = 2\alpha v_x = 2\alpha^2 y$$

$$\vec{F} = m\vec{a} = 2m\alpha^2(x\hat{x} + y\hat{y})$$

5. (*c*)



$$\frac{T}{\sqrt{2}} = 100$$

$$\frac{T}{\sqrt{2}} = F$$

$F = 100$ N

$\Rightarrow$ The graph given in option (*a*) suits the best for the above relation.

6. (*a*) Let additional force is $\vec{F}$, so that $F_{net} = 0$

$$\vec{F} + 5\left(+\hat{i}\right) + 6\left(-\hat{i}\right) + 7\left(+\hat{j}\right) + 8\left(-\hat{j}\right) = \vec{0}$$

$$\Rightarrow \vec{F} - \left(\hat{i} + \hat{j}\right) = \vec{0} \Rightarrow \vec{F} = \hat{i} + \hat{j}$$

$$\Rightarrow |\vec{F}| = \sqrt{1^2 + 1^2} = \sqrt{2}N$$

$$\theta = \tan^{-1}\left(\frac{\text{Component of } \hat{j}}{\text{Component of } \hat{i}}\right)$$

$$\theta = \tan^{-1}\left(\frac{1}{1}\right) = 45^o$$

7. (*c*) $Ma = T - Mg$ (While climbing up)

$T_1 = M(a + g)$

$T_1 = 50(5 + 10)$

$= 750$ N (Rope will break)

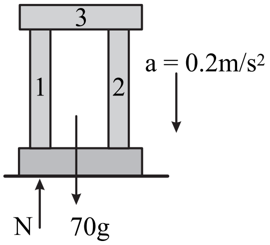
8. (*b*) From Newton's second law, $\vec{a} = \frac{\vec{F}}{m} = 8\hat{i} + 2\hat{j}$

From Kinematic equation, $\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2$

$\vec{r} = 0 + \frac{1}{2}\left(8\hat{i} + 2\hat{j}\right)10^2$

$\vec{r} = 400\hat{i} + 100\hat{j}$

9. (*d*) Using free body diagram given below.

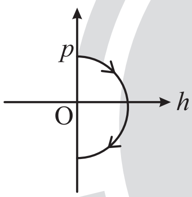


$70g - N = 70a = 70 \times 0.2 = 14$

$\Rightarrow N = 700 - 14 = 686\text{N}$

10. (*c*) Momentum $p = mv$ ...(*i*)

Motion under gravity $h = \dfrac{u^2 - v^2}{2g}$

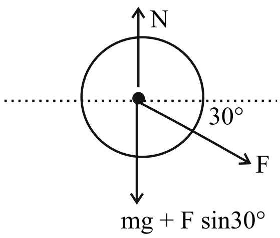


$h = \dfrac{u^2 - p^2/m}{2g}$ $(\because p = mv)$

11. (*c*) Equating the vertical forces $N = mg + F \sin 30^\circ$

$= 700 + 200 \times \frac{1}{2}$

$= 800\ N.$



12. (*b*) Impulse = Area under $F - t$ curve

(*a*) Impulse $= \frac{1}{2} \times 1 \times 0.5 = \frac{1}{4} N.s$

(*b*) Impulse = 0.5 × 2 = 1 N.s (maximum)

(*c*) Impulse $= \frac{1}{2} \times 1 \times 0.75 = \frac{3}{8} N.s$

(*d*) Impulse $= \frac{1}{2} \times 2 \times 0.5 = \frac{1}{2}$ N.s

13. (*c*) $\left|\dfrac{d\vec{p}}{dt}\right| = \left|\vec{F}\right| \Rightarrow \dfrac{d\vec{p}}{dt} =$ Slope of curve

max slope (*c*), ⇒ min slope (*b*)

14. (*b*) $P_i = Nmv\hat{i}$ and $\vec{P}_f = -Nmv\hat{i}$

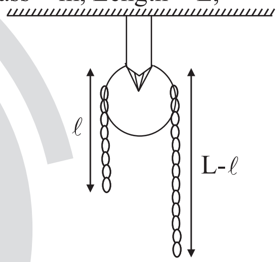
where N is the number of balls that strike the wall

$\Delta\vec{P} = \vec{P}_f - \vec{P}_i = -2Nmv\hat{i} = -200\,Nmv\hat{i}$

$\vec{F}_{Total} = \dfrac{\Delta\vec{P}}{\Delta t} = -\dfrac{200mvt}{t}$

$\left|\vec{F}\right| = \dfrac{200mv}{t}$

15. (*d*) Let, Mass = m, Length = L,



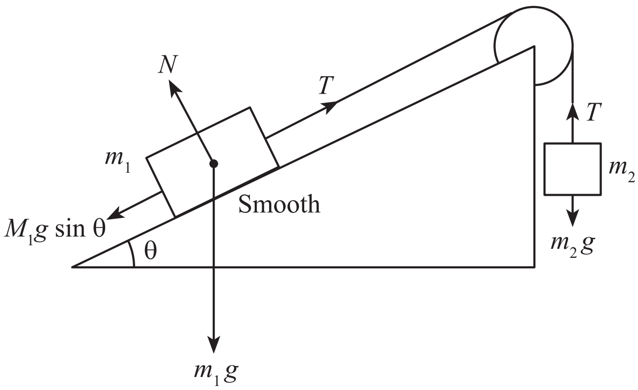
$\dfrac{m}{L} = \lambda$

$a = \dfrac{F_{\text{Pulling}}}{\text{mass}} = \dfrac{(L-\ell)\lambda g - \lambda\ell g}{\lambda L} = \dfrac{(L-2\ell)\lambda g}{\lambda L} = \dfrac{g}{2}$

$\Rightarrow L - 2\ell = \dfrac{L}{2};\ \ell = \dfrac{L}{4} \Rightarrow x = 4$

16. (*b*) As m_1 is in rest

$T = m_2 g = 30\ N$



$T = m_1 g \sin\theta \Rightarrow 50 \sin\theta = 30$

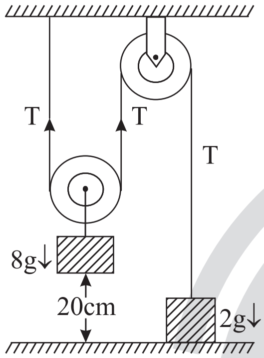
$$\Rightarrow \sin\theta = \frac{3}{5} \Rightarrow \theta = 37°$$

$$\therefore N = 50\cos\theta = 50\cos 37° = 50\times\frac{4}{5} = 40N$$

17. (*b*) $a_1 = \dfrac{2M_1g - M_1g}{2M_1 + M_1} = \dfrac{g}{3}$

$a_2 = \dfrac{3M_1g - M_1g}{3M_1 + M_1} = \dfrac{g}{2} \;\therefore\; \dfrac{a_1}{a_2} = \dfrac{2}{3}$

18. (*b*) From the free body diagram,



For 8 kg block,

$8g - 2T = 8a$ ...(*i*)

For 2 kg block,

$T - 2g = 2\times 2a$ ...(*ii*)

Solving equations (*i*) and (*ii*)

$$a = \frac{g}{4} = 2.5\text{m/s}^2$$

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2\times 0.2}{2.5}} = 0.4s$$

19. (*b*) For upward motion

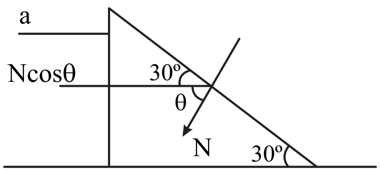
v = constant

$F = (+2\text{N})$

For downward motion

$F = (-2\text{N})$

20. (*c*) $N\cos 60° = Ma = 16a \Rightarrow N = 32a$



For block w.r.t wedge ⊥ to incline

$N = 8g\cos 30° - 8a\sin 30°$

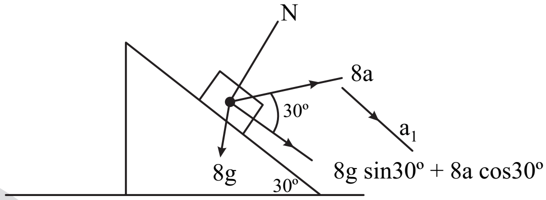
$\Rightarrow 32a = 8g\cos 30° - 8a\sin 30°$

$$\Rightarrow 32a = 4\sqrt{3}g - 4a$$

$$\Rightarrow 36a = 4\sqrt{3}g$$

$$\Rightarrow a = \frac{\sqrt{3}}{9}g$$

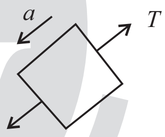
Now for 8 kg, along incline,



$ma_1 = 8g\sin 30° + 8a\cos 30°$

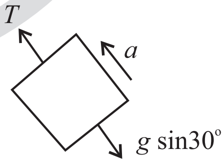
$$a_1 = g\times\frac{1}{2} + \frac{\sqrt{3}}{9}g\times\frac{\sqrt{3}}{2} = \frac{g}{2} + \frac{3}{18}g = \frac{2}{3}g$$

21. (*b*) Free body diagram for the 4 kg block



$4g\sin 60° - T = 4a$ ...(*i*)

For the 1 kg block,



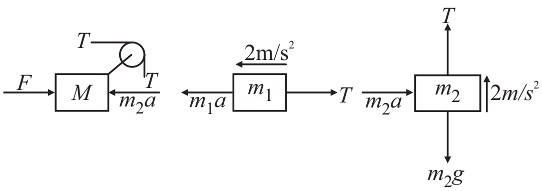
$T - g\sin 30° = a$ ...(*ii*)

Solving equation (*i*) and (*ii*)we get.

$$20\sqrt{3} - T = 4T - 20$$

$$T = 4\left(\sqrt{3}+1\right)N$$

22. (*a*)



For m_2

$20 \times 2 = T - 200$

$T = 240$ N

For m_1

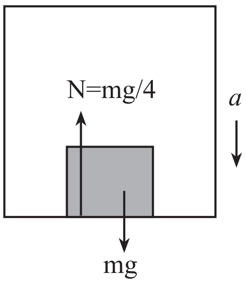
$10 \times 2 = 10a - T$

$a = 26$ m/s^2

For M

$$Ma = (M + m_2)a + T$$
$$= 120 \times 26 + 240$$
$$= 3360 \text{ N}$$

23. (*c*) FBD of m

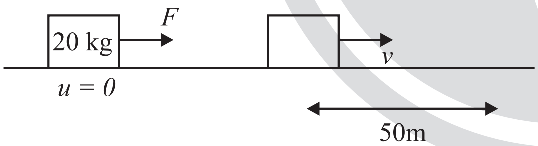


$$W' = m(g-a)$$

$$W' = mg\left(1-\frac{a}{g}\right) = W\left(1-\frac{a}{g}\right)$$

$$\frac{W}{4} = W\left[1-\frac{a}{g}\right] \Rightarrow a = \frac{3g}{4}$$

24. (*b*)



$50 = v \times 10$

$v = 5\ m/s$ $\quad [\because v = u + at]$

$\Rightarrow v = 0 + a \times 20$

$$a = \frac{1}{4} m/s^2$$

$$F = ma = 20 \times \frac{1}{4} = 5\,N$$

25. [12] $\vec{F} = \left(2\hat{i} + 3\hat{j} + 5\hat{k}\right)N$ time = 4 *sec* As body start from rest therefore position vector initially $\vec{r}_i = \left(0\hat{i} + 0\hat{j} + 0\hat{k}\right)$ and u (initial velocity) = 0 given, $r_f = \left(x\hat{i} + y\hat{j} + z\hat{k}\right)$ position vector initially $\vec{r}_i = \left(0\hat{i} + 0\hat{j} + 0\hat{k}\right)$ u (initial velocity) = 0 given,

$r_f = \left(x\hat{i} + y\hat{j} + z\hat{k}\right)$

Now, from second equation of motion

$$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2 \Rightarrow r_f - r_i = \frac{1}{2} \times \left(\frac{2\hat{i} + 3\hat{j} + 5\hat{k}}{2}\right) \times (4)^2$$

$$\Rightarrow (x\hat{i} + y\hat{j} + z\hat{k}) - (0\hat{i} + 0\hat{j} + 0\hat{k}) = 8\hat{i} + 12\hat{j} + 20\hat{k}$$

$$\Rightarrow x\hat{i} + y\hat{j} + z\hat{k} = 8\hat{i} + 12\hat{j} + 20\hat{k}$$

$\therefore$ The value of $b = 12$

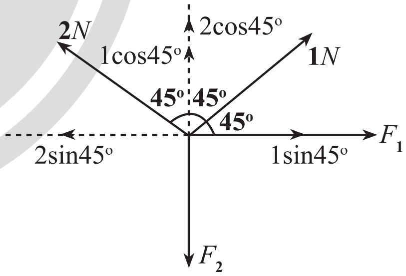
26. [3] $F_2 = 1\cos45^\circ + 2\cos45^\circ = 3\cos45^\circ = \dfrac{3}{\sqrt{2}}$ N

$$F_1 + 1\sin45^\circ = 2\sin45^\circ$$

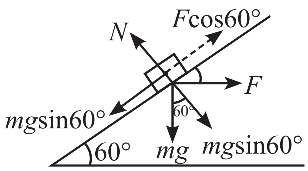
$$F_1 = 1\sin45^\circ = \frac{1}{\sqrt{2}} \text{ N}$$

$$\frac{F_1}{F_2} = 1:3$$

$x = 3$



27. [12] For equilibrium of block along the inclined plane

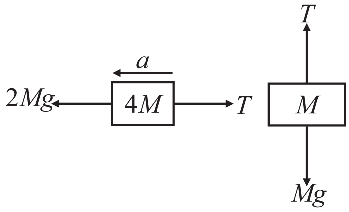


$$F\cos60^\circ = mg\sin60^\circ$$

$$F = 0.2 \times 10 \times \frac{\sqrt{3}}{2} \times \frac{2}{1} \Rightarrow F = \sqrt{12} \text{ N}$$

$\Rightarrow x = 12$

28. [6]



for M

$Ma = T - Mg$ ...(*i*)

for 4 M

$4Ma = 2Mg - T$ ...(*ii*)

From (*i*) and (*ii*)

$a = \frac{g}{5}$ and $T = \frac{6}{5}Mg$

So $x = 6$

29. [164] Suppose that the magnitude of components along AB and AC are F_1 and F_2 respectively.

According to the question, $\vec{P} + \vec{F_1} + \vec{F_2} = \vec{0}$

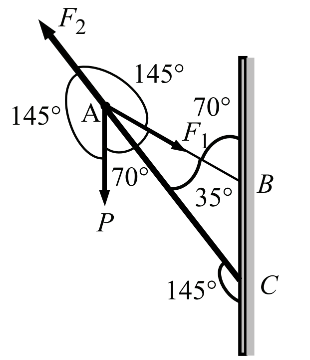
By using sign Rule, we can write

$$\frac{F_1}{\sin(145^0)} = \frac{F_2}{\sin(70^0)} = \frac{P}{\sin(145^0)}$$

$$\Rightarrow F_2 = \frac{P\sin(180^0 - 110^0)}{\sin(110^0 + 35^0)}$$

$$\Rightarrow F_2 = \frac{P\sin(110^0)}{\sin(110^0)\cos(35^0) + \cos(110^0)\sin(35^0)}$$

$$= \frac{100 \times 0.939}{0.939 \times 0.819 - 0.342 \times 0.573} \approx 164N$$



Hence, component of force along AC is $-F_2 = -164$ N

30. [492] Net accentuation in vertical direction is a,

$mg - N = ma$

$N = m(g - a) = 60 \times (10 - 1.8) = 60 \times 8.2 = 492\ N$